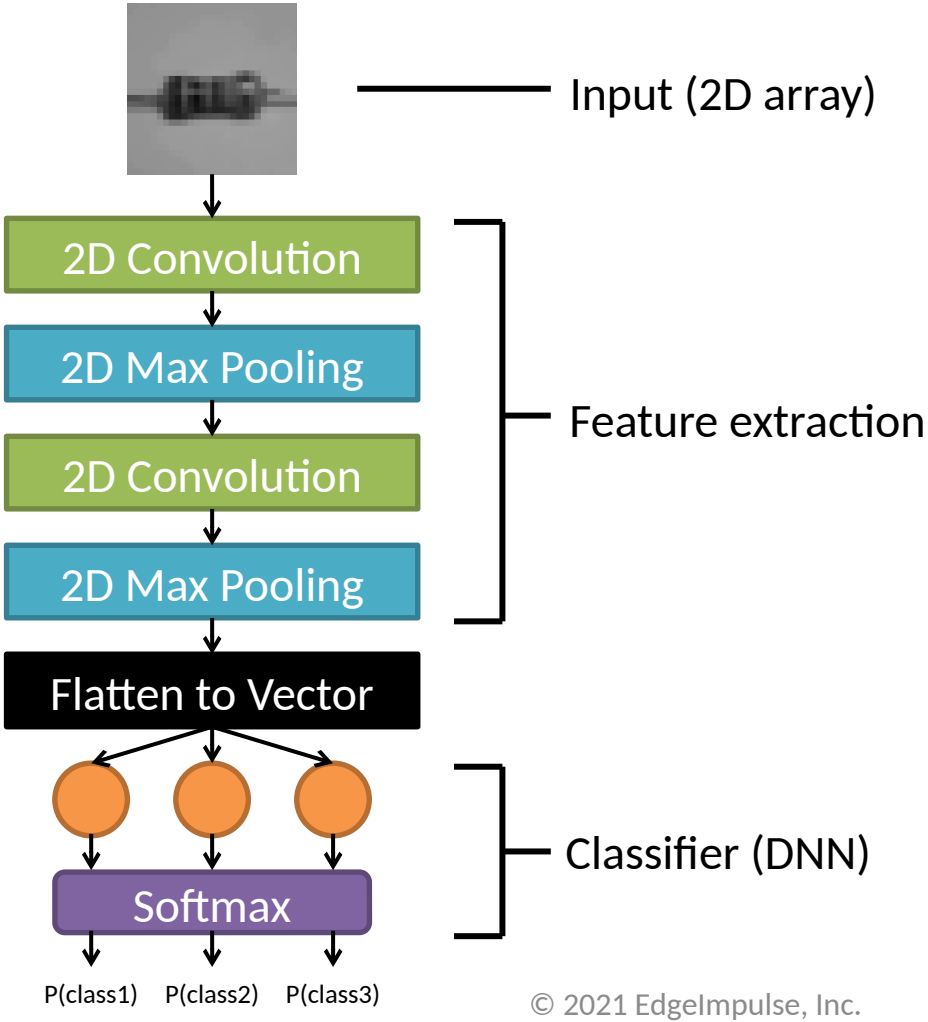
Computer Vision with Embedded Machine Learning

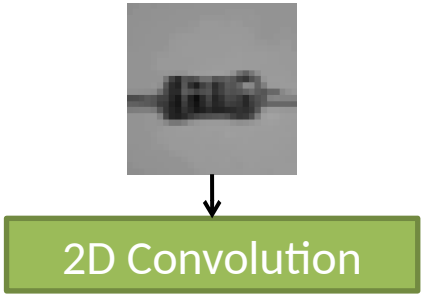
Convolutional Neural Network

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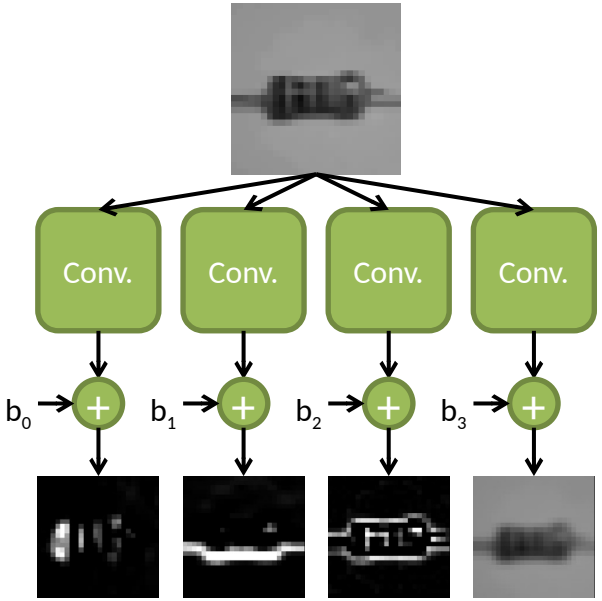


Popular input formats:
- Float [0.0..1.0]
- Uint8 [0..255]
- Int8 [-128..127]

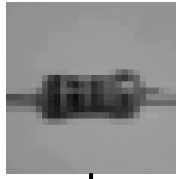
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Number of filters: 4
Kernel: 3x3
Stride: 1
Padding: Same



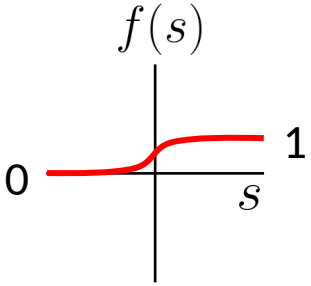
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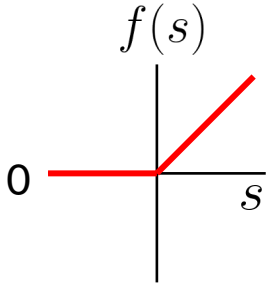
2D Convolution

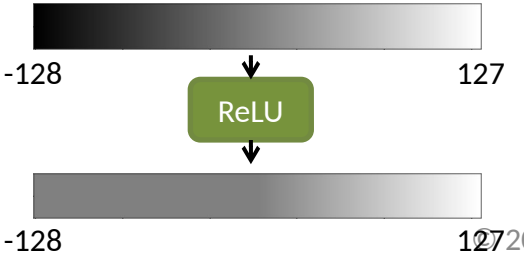
Number of filters: 4
Kernel: 3x3
Stride: 1
Padding: Same

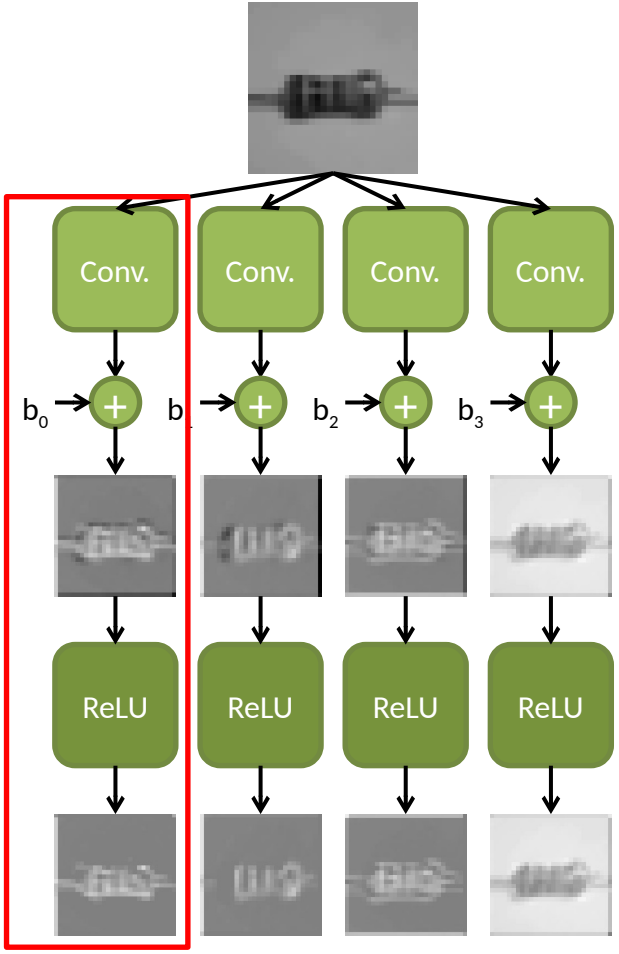
Sigmoid function



ReLU function

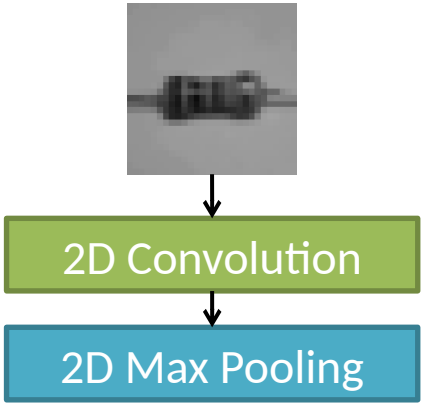




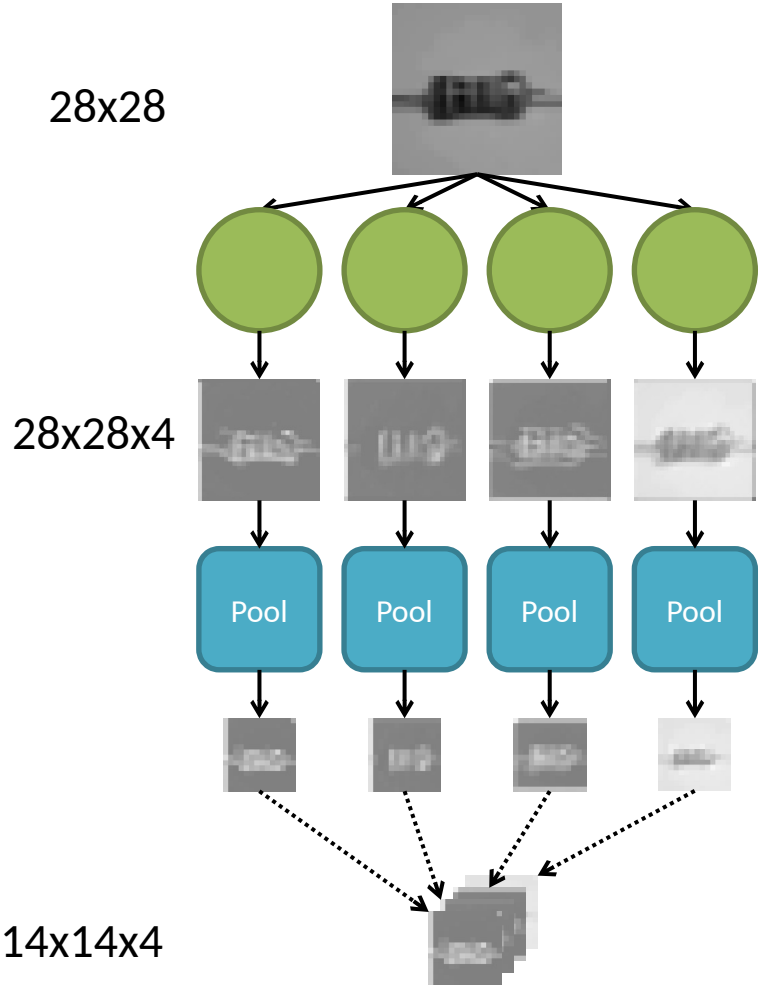


"neuron" or "node"

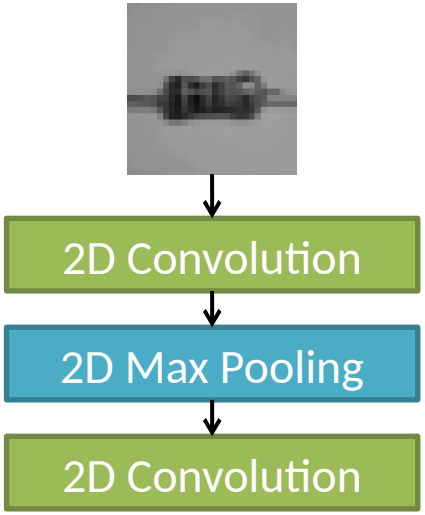
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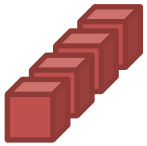
Shape: 2x2
Stride: 2



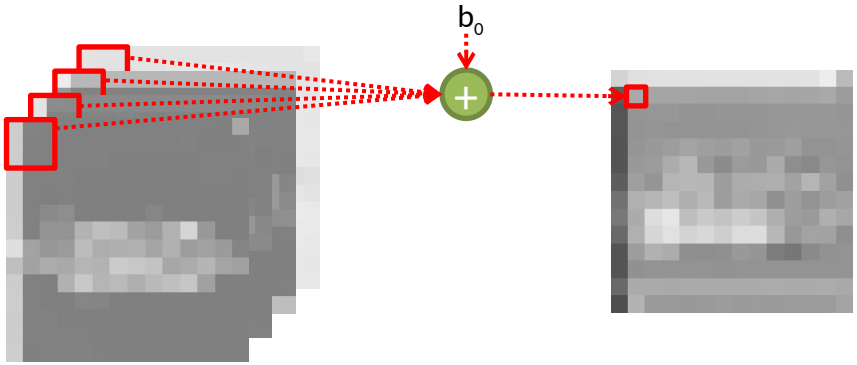
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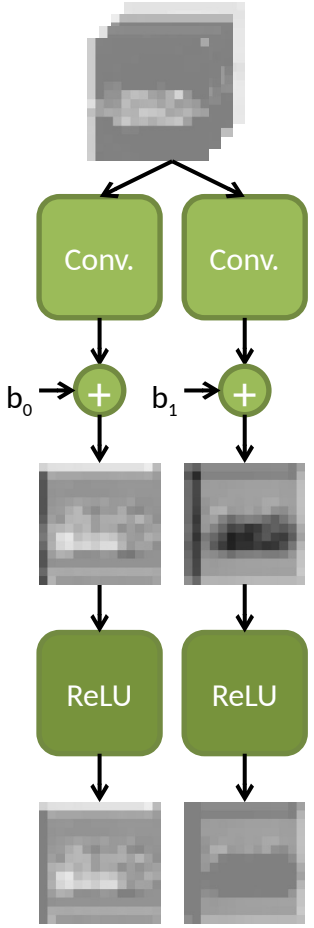


Number of filters: 2
Kernel: 3x3
Stride: 1
Padding: Same



Kernel: 3x3x4

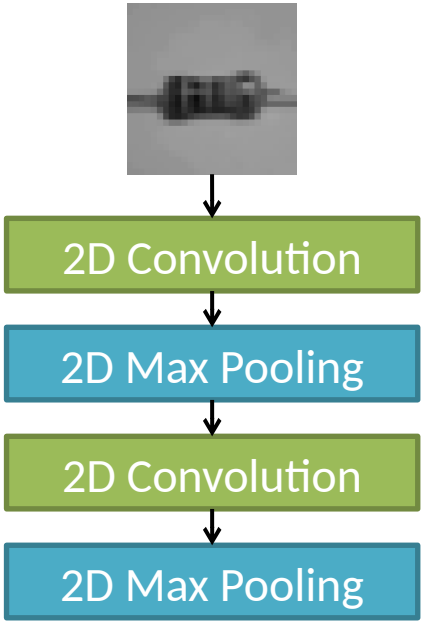




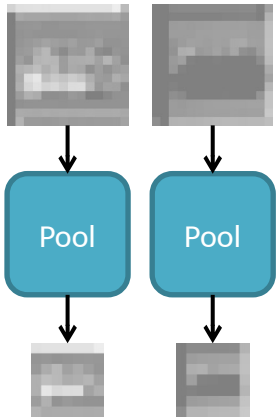
14x14x4

14x14x2

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Shape: 2x2
Stride: 2

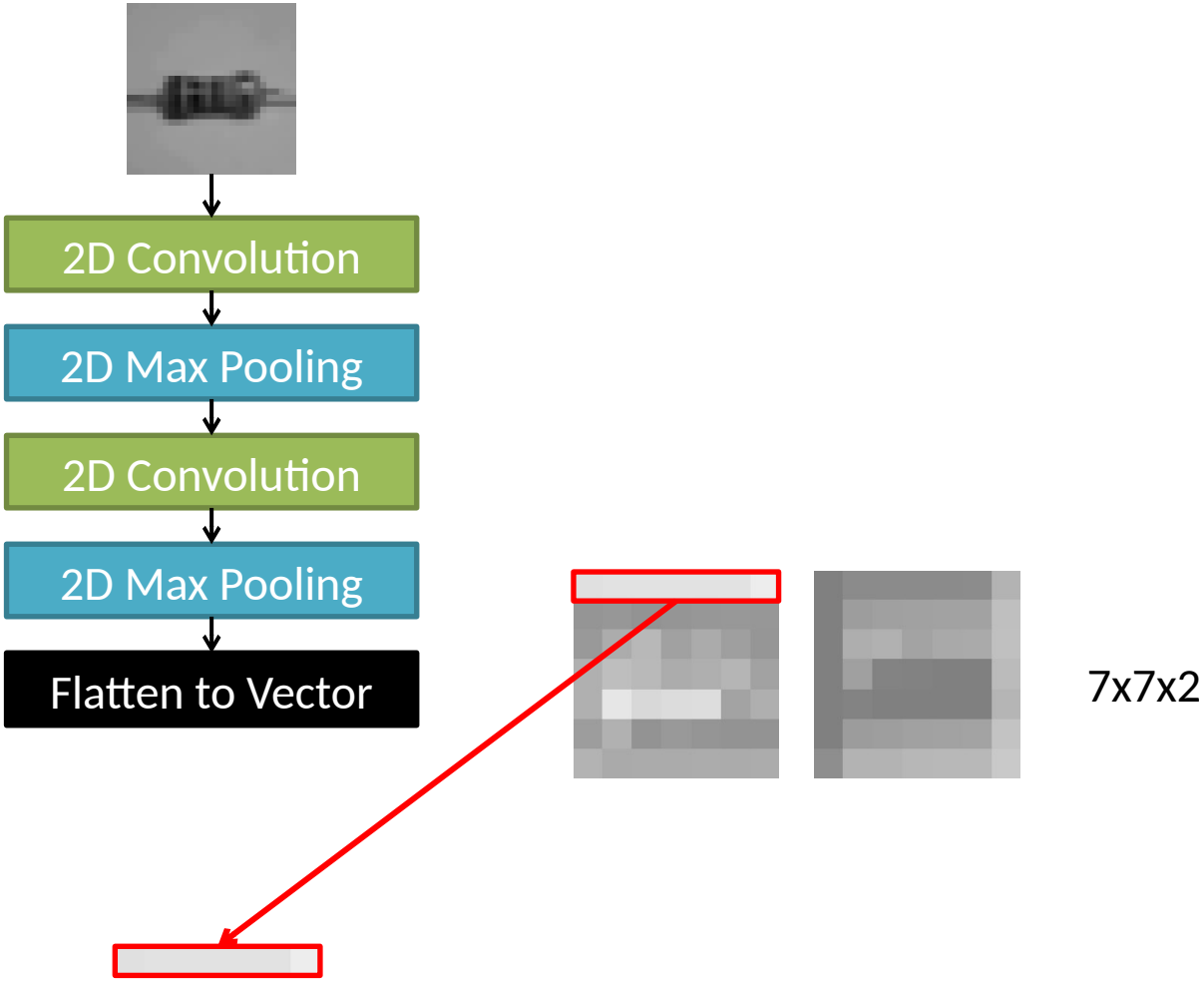


14x14x2

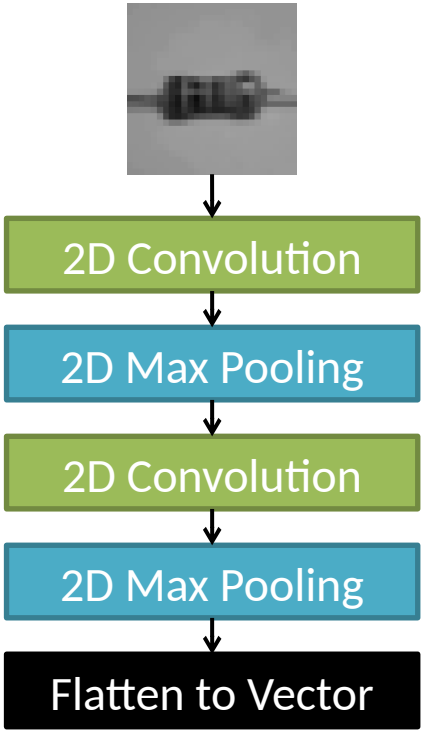


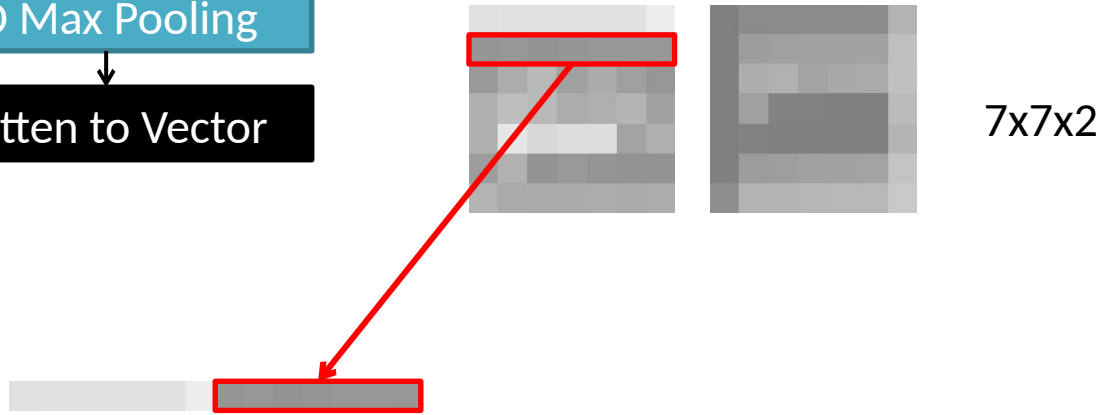
7x7x2

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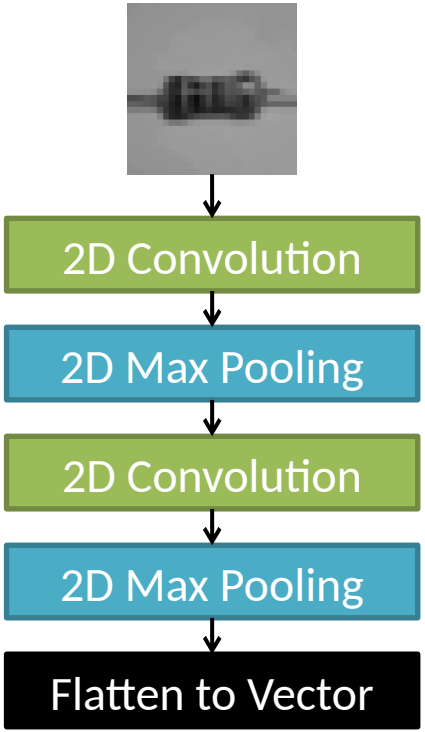


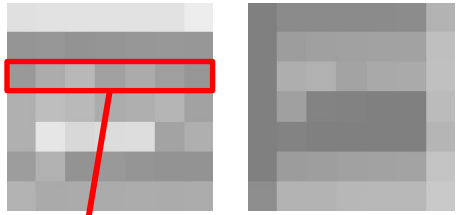
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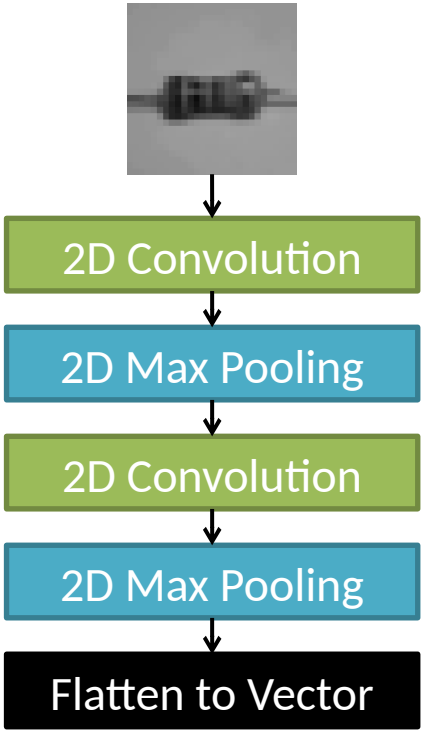


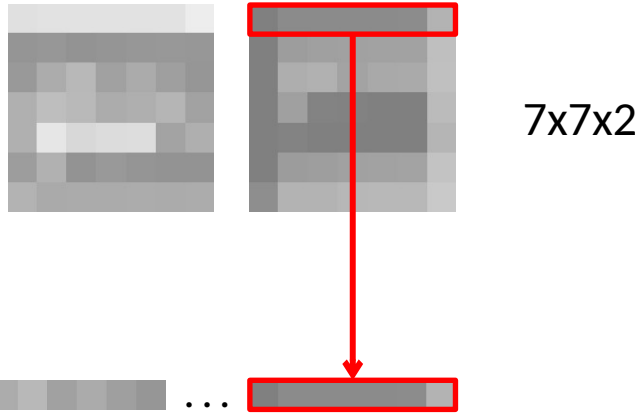


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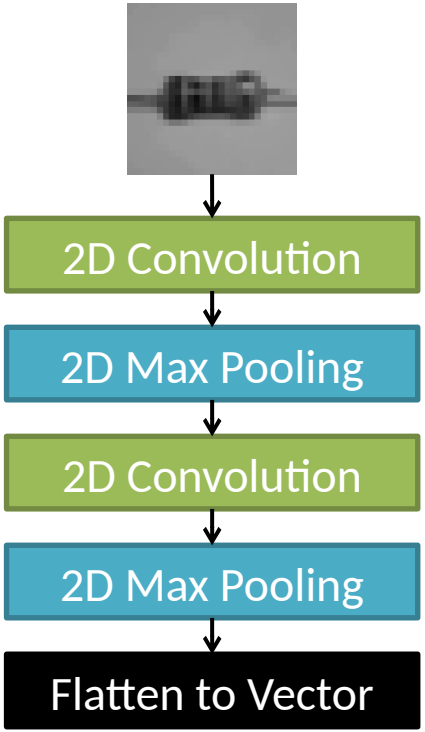


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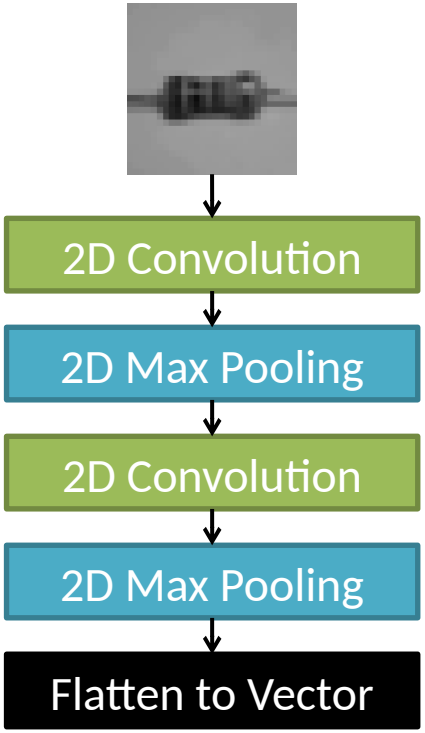


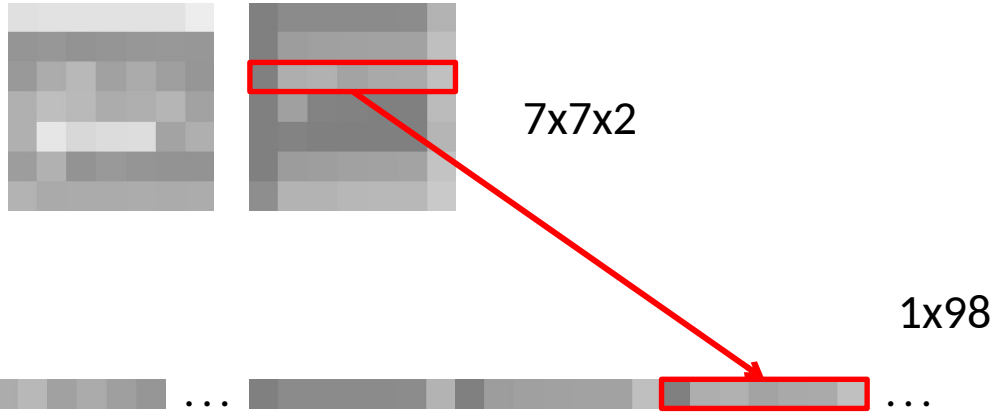
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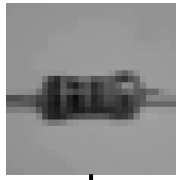


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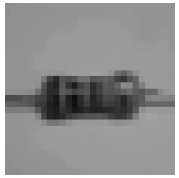
2D Convolution

2D Max Pooling

2D Convolution

2D Max Pooling

Flatten to Vector

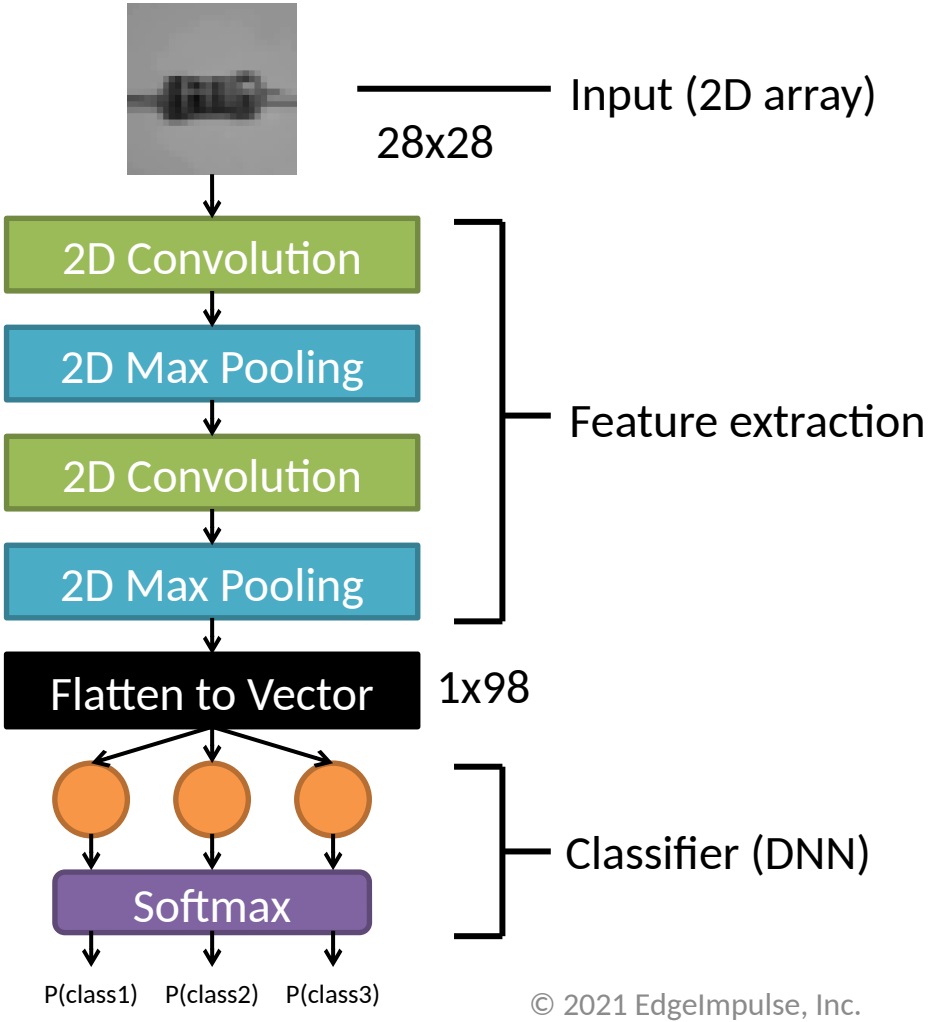


Flatten original image: 28x28 = 784

1x98



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